
Notes for ‘Increasing and decreasing functions’

Important Ideas and Useful Facts:

- (i) **Increasing functions:** A function $y = f(x)$ is *increasing* if the y -values increase as the x -values increase. Here, when we say that *values increase*, we are using the technical meaning that values move to the right along the x -axis or upwards along the y -axis, in each case, moving in the positive direction (regardless of the magnitude of the values).

If the function is increasing and its graph is a smooth curve, then the graph is sloping upwards as one moves from left to right.

Recall that relative position to the left or right of the real number line is captured by the $<$ relation. More precisely, we say that f is *increasing* if, for any a and b in the domain of f , we have

$$a < b \implies f(a) < f(b) .$$

Because this property is so important in applications, it is common to look for intervals on the real line where the function becomes increasing, by restricting its domain to that interval.

Clearly, the identity function $y = x$ is increasing. More generally, any line with positive slope is increasing.

The square function $y = x^2$ is not increasing over $\mathbb{R}$, but becomes increasing, for example, if the domain is restricted to $[0, \infty)$.

The cube function $y = x^3$ is increasing on all of $\mathbb{R}$.

- (ii) **Decreasing functions:** A function $y = f(x)$ is *decreasing* if the y -values decrease as the x -values increase. Here, when we say that *values decrease*, we are using the technical meaning that values move to the left along the x -axis or downwards along the y -axis, in each case, moving in the negative direction (regardless of the magnitudes of the values).

If the function is decreasing and its graph is a smooth curve, then the graph is sloping downwards as one moves from left to right.

A precise definition, using the $<$ relation, is that f is *decreasing* if, for any a and b in the domain of f , we have

$$a < b \implies f(a) > f(b) .$$

Clearly, the function $y = -x$ is decreasing. More generally, any line with negative slope is decreasing.

The square function $y = x^2$ is not decreasing over $\mathbb{R}$, but becomes decreasing, for example, if the domain is restricted to $(-\infty, 0]$.

The negative cube function $y = -x^3$ is decreasing on all of $\mathbb{R}$.

- (iii) **Invertibility:** An increasing function is invertible and its inverse is also increasing.

For example, the natural exponential function is increasing and its inverse, the natural logarithm function is also increasing, though its domain is restricted to the positive half of the real line $(0, \infty)$.

The restriction of the sine function $y = \sin x$ to the interval $[-\frac{\pi}{2}, \frac{\pi}{2}]$ is increasing, and its inverse, the inverse sine function $y = \sin^{-1} x$ is also increasing, with domain $[-1, 1]$.

The restriction of the tan function $y = \tan x$ to the interval $(-\frac{\pi}{2}, \frac{\pi}{2})$ is increasing, and its inverse, the inverse tan function $y = \tan^{-1} x$ is also increasing, with domain all of $\mathbb{R}$.

A decreasing function is invertible and its inverse is also decreasing.

For example, the restriction of the cosine function $y = \cos x$ to the interval $[0, \pi]$ is decreasing, and its inverse, the inverse cosine function $y = \cos^{-1} x$ is also decreasing, with domain $[-1, 1]$.

- (iv) **Reflection in the y -axis:** If the graph of an increasing function is reflected in the y -axis then we obtain the graph of a decreasing function, and vice-versa.

For example, $y = e^x$ is increasing (and represents exponential growth). The result of reflecting its graph in the y -axis produces the function $y = e^{-x}$, which is decreasing (and represents exponential decay).

- (v) **Sign of the derivative:** Consider a function $y = f(x)$. If the derivative of f is positive over some interval, then the function is increasing over that interval.

This makes sense intuitively. If, for example, y represents displacement of a car travelling along a straight road at time x , and the derivative is positive, then the velocity is positive and the car is always moving in the positive direction, corresponding to the function being increasing.

If the derivative of f is negative over some interval, then the function is decreasing over that interval.

Again, the car analogy is useful. If y represents the displacement of a car at time x and the derivative is negative then the car would have negative velocity and be moving in reverse, that is, in the negative direction, corresponding to the function being decreasing.

Examples and derivations:

1. We explain why reflecting an increasing function in the y -axis produces a decreasing function. Suppose $y = f(x)$ is increasing over some domain. Let $y = g(x)$ be the function corresponding to reflecting the graph of $y = f(x)$ in the y -axis, so that

$$g(x) = f(-x)$$

for any input x for g . Suppose that a and b are inputs for g and $a < b$. Then $-a$ and $-b$ are inputs for f and $-b < -a$, since multiplying by -1 reverses the inequality. Hence, we have

$$g(b) = f(-b) < f(-a) = g(a),$$

since f is increasing. Thus $g(a) > g(b)$, completing the verification that g is decreasing.

A similar argument shows that reflecting a decreasing function in the y -axis produces an increasing function.

2. We explain why an increasing function is invertible and why its inverse function must also be increasing. Suppose that $y = f(x)$ is increasing. If the horizontal line test fails then there would be two inputs, say a and b such that $a < b$ and $f(a) = f(b)$, which contradicts that $f(a) < f(b)$. Hence the horizontal line test is satisfied and f is invertible.

We show that the inverse function f^{-1} is also increasing. Suppose c and d are inputs for f^{-1} such that $c < d$. Our task is to show that $f^{-1}(c) < f^{-1}(d)$. Put

$$a = f^{-1}(c) \quad \text{and} \quad b = f^{-1}(d) ,$$

so that $f(a) = c$ and $f(b) = d$, by definition of the inverse function. If $a = b$ then $c = f(a) = f(b) = d$, contradicting that $c < d$. Hence $a \neq b$. If $b < a$ then $f(b) < f(a)$, since f is increasing, so that $d = f(b) < f(a) = c$, again contradicting that $c < d$. The only alternative is that $a < b$, so that

$$f^{-1}(c) = a < b = f^{-1}(d) ,$$

completing the verification that f^{-1} is increasing.

A similar argument shows that a decreasing function is invertible and its inverse is also decreasing. This also follows from the result about increasing functions, by reflecting a given decreasing function in the y -axis, using the fact that the reflected function is now increasing with an increasing inverse function; this new inverse function, in turn, reflects in the x -axis, which it is easy to check produces a decreasing inverse function corresponding to the original decreasing function.

3. Consider the function $y = f(x) = x^3 - 12x$. Then

$$y' = 3x^2 - 12 = 3(x^2 - 4) = 3(x+2)(x-2) ,$$

which is positive if $x > 2$ and if $x < -2$, and negative if $-2 < x < 2$. Thus f is increasing for $x < -2$, becomes decreasing for $-2 < x < 2$ and then becomes increasing again for $x > 2$. This increasing-decreasing-increasing behaviour can be seen on the graph of f below. Soon you will develop techniques to understand fully how to obtain such a graph and all of its important features.

